# Green Score — algorithms, sources, and an audit

**From:** Engineering

**Date:** 2026-07-27 (revised; first issued 2026-07-25)

**Subject:** What the Green Score computes, where every number comes from, and how accurate it is

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## Summary

This sets out the Green Score end to end: the algorithm, where every constant comes from, what the model actually produces, and — new in this revision — **how accurate it is against satellite measurement**, in both senses: how close the ward-level temperature is (§5A), and whether the pattern *within* a ward is real (§5A, "What those figures do NOT cover"). The second answer is largely no, it is disclosed in the product, and it matters more than the first for anyone reading the map block by block. Section 8 lists every source with a resolvable link.

Writing it prompted a proper review of the constants, which was overdue. That turned up **five** that were uncited, self-referential, or not scientifically supportable — all now corrected and sourced (§4). A validation harness scores model output against published measurements on every run, so the model can't drift quietly again. It passes **8 of 8** checks, up from 4 of 10.

**What changed since the first issue.** The previous version said plainly that nothing had been validated against measurement for our own wards. That is no longer true. We built a calibration set of **49 usable NASA ECOSTRESS scenes** over Kolkata (2024-01 → 2026-07, 179 acquisitions attempted, every exclusion recorded), attached hour-matched meteorology to each, and compared the model to observation scene by scene. Section 5A reports what that found, including the parts that did not go our way.

The headline result is a number we can now put on the tool:

| | measured accuracy | status in the interface |
|---|---|---|
| **Night** | ± **3.5 °C** (20 scenes) | labelled *calibrated* |
| **Day** | ± **5.0 °C** (12 scenes) | labelled *indicative only* |

Daytime is worse for a reason that no amount of engineering will fix, and §5A explains it: the limit is the resolution of the weather data driving the model, not the physics inside it. Both figures are now shown next to the temperature readout on the tool itself, rather than living only in this document.

Two earlier findings are worth reading before the detail. The headline heat figure looked roughly eight times too high against published Kolkata values and was initially recorded as our most serious defect; it turned out to be **mislabelled rather than miscalibrated** (§4.1). And the attempt to fit the model's remaining free constants to measurement **failed its acceptance test** — reported in §5A rather than quietly retried, because the failure is informative.

Section 6 sets out what I still don't trust, without hedging.

## 1. What the Green Score computes

A 0–100 number, the equal-weighted mean of three sub-scores:

```
Green Score = 100 × [ greening ratio + cooling achieved + budget efficiency ] ÷ 3
```

Each sub-score is capped at 1.0, so the total cannot be bought with spending. **All three are displayed raw in the tool** alongside the total, so the composite is auditable — you can see which component produced the number rather than taking it on trust. Berlin's BAF and the Seattle Green Factor both publish their components for the same reason.

### Greening ratio

A **weighted-area ratio** — the same structure used by every established green-space planning standard: Berlin's **Biotope Area Factor** (mandatory since 1994), the **Seattle Green Factor**, and Singapore's **Green Plot Ratio**. All compute *sum of (area × ecological weight) ÷ total area*.

| Surface | Weight | Basis |
|---|---|---|
| In-ground vegetation | 1.0 | Berlin BAF "vegetation connected to soil" |
| Tree canopy over street | 0.6 | Berlin / Seattle canopy bands |
| Green facade | 0.6 | Berlin wall greening 0.5; Seattle up to 0.7 |
| Water / wetland | 0.8 | Berlin water surface |
| Cool roof | 0.1 | Our own low weight — a painted roof is not habitat |
| Sealed surface | 0.0 | All three standards |

Normalised against **0.45**, the midpoint of Berlin's 0.30–0.60 target band.

### Cooling achieved

Modelled surface-temperature drop versus the no-intervention baseline, normalised against 2.5 °C.

### Budget efficiency

Degrees cooled per ₹ crore, normalised against 0.15 °C/crore.

## 2. Where every number comes from

**Measured** = from data · **Cited** = traceable to a named source · **Tool-relative** = normalised against our own output · **Placeholder** = explicitly temporary.

### Costs

| Item | Value | Grade | Source |
|---|---|---|---|
| Cool roof | ₹150/m² | Cited — high | Telangana Cool Roof Policy 2023-28; NRDC India |
| Street tree | ₹1,500/tree | Cited — med-high | Municipal planting + 3-yr maintenance |
| Pocket park | ₹1.5 crore/ha | Cited — medium | Gujarat AMRUT 2.0 (2 named projects) |
| Green facade | ₹9,500/m² | Cited — medium | Indian vendor pricing; excludes irrigation opex |
| Wetland | *no figure exists* | **Open** | See §6 |

### Physical coefficients

| Item | Value | Grade | Source |
|---|---|---|---|
| Dark roof albedo | 0.15 | Cited — high | LBNL Heat Island Group |
| Aged cool-roof albedo | 0.60 | Cited — high | LBNL *aged* value, not fresh-white 0.85 — deliberately conservative |
| Pocket park radius | 50 m (0.785 ha) | Cited — **Kolkata-specific** | Mitra et al. 2022, threshold value of efficiency |
| **Evapotranspiration L** | **0.43** | **Derived — see §4.2; contested by satellite fit, see §5A** | Solved against two independent constraints |
| **Facade heat reduction** | **0.03** | **Cited — corrected from 0.30** | Gunawardena & Steemers 2023 |
| Influence kernel λ | ≈47 m | Empirical, honestly labelled | See §4.4 |
| **Sky temperature** | **computed, not fixed** | **Cited** | Brutsaert (1975) clear-sky emissivity, cloud-screened; replaced two hard-coded values |
| **Night evapotranspiration** | **10 % of daytime** | **Cited** | Stomata close after dark; tapers to zero below the dewpoint |
| **Night anthropogenic heat** | **50 % of daytime** | **Estimated, bounded** | Diurnal load profiles, South and East Asian cities |
| **Warming pathways** | **+1.25 / +4.1 °C** | **Cited — corrected** | Dhara et al. 2025, PLOS Climate |

### Score parameters

| Item | Value | Grade |
|---|---|---|
| **Weights** | **equal thirds** | **Corrected from 40/40/20 — see §4.5** |
| Greening reference | 0.45 | Cited, but transplanted (see §6) |
| Cooling reference | 2.5 °C | Cited-ish |
| Efficiency reference | 0.15 °C/cr | **Tool-relative — labelled, not fixed** |

### Input geometry — genuinely measured

Microsoft ML Building Footprints (ODbL) · Google Open Buildings 2.5D heights (CC BY 4.0) · OpenStreetMap road centrelines (ODbL) · Norwegian Meteorological Institute live air temperature

Added for validation (§5A): NASA ECOSTRESS 70 m land-surface temperature (public domain) · NASA POWER hourly meteorology (public domain) · ESA WorldCover 10 m land cover (CC BY 4.0) · GHS-SMOD R2023A settlement classification (CC BY 4.0). All four are free and redistributable; none carries a non-commercial clause. Open-Meteo was the easier route for historical weather and was **rejected on licence grounds**, being non-commercial.

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## 3. What the model produces

| Scenario | Cooling | Cost | Score |
|---|---|---|---|
| Nothing | 0.00 °C | ₹0 | 2 |
| Cool roofs 100% | 2.25 °C | ₹7.5 cr | 69 |
| Tree corridors max | 2.01 °C | ₹66 lakh | **84** |
| Green facades max | **0.14 °C** | **₹190 cr** | **4** |
| Everything | 4.45 °C | ₹203 cr | 66 |

**Trees score highest, and facades now score almost nothing.** Both are consequences of the corrections, not of design. Cool roofs cost **₹3.3 crore per °C**; facades cost **₹1,382 crore per °C**. That ordering matches Indian Heat Action Plan practice, where cool roofs lead — and it emerges from cited costs rather than being put there.

**Doing everything scores lower than trees alone (66 vs 84).** Deliberate: the cooling term caps at 2.5 °C, so extra cooling earns nothing while cost destroys the efficiency term. It rewards cost-effectiveness. It is counterintuitive enough that the interface should explain it, and currently doesn't.

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## 4. The audit — five corrections

### 4.1 The heat figure was mislabelled, not miscalibrated

The tool displayed "**UHI Δ vs rural: +11 °C**". Published surface UHI for Kolkata is **0.85–1.5 °C** (Nayak 2023, Jain 2023, Siddiqui 2021). That looks like an eight-fold overstatement, and I initially wrote it up as our most serious defect.

**It isn't.** Voogt & Oke (2003), the canonical reference, put clear-sky **daytime surface** UHI at **10–15 °C**, while canopy-layer **air** UHI is 2–5 °C. The published Kolkata figures are 1 km MODIS, season-averaged, measured against a real rural population. Ours is clear-sky midday at 7 m against a synthetic all-vegetation cell. **Three different quantities.**

The number is defensible. The label was not: "UHI" and "rural" are both load-bearing terms we had no claim to.

**Fixed:** renamed to "**Δ vs all-green ref**", with the reference stated in the tooltip and an explicit note that measured Kolkata SUHII is a different quantity.

I'm flagging this prominently because it is the error I would most want caught in my own work: I compared two numbers that shared a name and not a definition.

### 4.2 Evapotranspiration was over-strong — now derived, not tuned

The ET coefficient was 0.50, which put a fully-vegetated surface **5.8 K below air temperature** — more cooling than evapotranspiration can physically deliver.

Rather than pick a nicer number, I solved for the value satisfying **two independent published constraints at once**:

- park cool-island intensity of **4.83–8.07 °C** (Mitra et al. 2022, measured in Kolkata)
- vegetated surface no more than **4 K below air** (physical ceiling on ET)

They intersect at **0.40–0.46**. The value is now **0.43**, the midpoint. Both constraints now pass.

### 4.3 Green facades were overstated roughly tenfold

The facade heat-reduction coefficient was **0.30** with no source. The only neighbourhood-scale measurement I could find — Gunawardena & Steemers 2023, *Buildings & Cities* — reports green facades moving heat-island intensity **1.86 K → 1.81 K, about 3%**, and cutting space-conditioning energy 2.1%. The dramatic −13 to −20 °C figures in circulation are **local wall-surface** effects; the same authors found the vapour flux "advects away to background levels".

**Fixed:** coefficient **0.30 → 0.03**, cited. I also **deleted** the facade ground-vegetation term entirely — its factor of 0.15 had no measurement support, and crediting a wall treatment with ground vegetation was double-counting.

Consequence: facades went from an apparently competitive lever (1.90 °C) to a marginal one (0.14 °C). That is what the literature says they are.

### 4.4 A physical constant was tuned for appearance — now honestly named

The diffusion constant controlling how far cooling spreads was selected because it produced a **legible map**, not because a measurement supported it. Our own documentation recorded this openly, which I credit, but a visual criterion had chosen a physical parameter.

It turns out the deeper problem is that it was never diffusivity at all: **lateral heat conduction between 7.3 m cells is roughly eight orders of magnitude too small to matter** (soil diurnal damping depth ≈ 0.12 m). Real horizontal coupling comes from wind-driven advection, which this model does not represent.

**Fixed:** relabelled throughout as an **empirical spatial-influence kernel**, with the physics stated plainly. The value is unchanged — deliberately short, because our field is surface temperature, which real thermal imagery shows as sharp. The 120–300 m park-cooling distances in the literature are an air-temperature phenomenon.

### 4.5 Two pathway figures were scientifically indefensible

The projection slider offered **−1.2 °C ("target 2030")** and **+2.4 °C ("BAU 2040")**, both uncited.

**No emissions scenario produces regional cooling over India.** Even the most aggressive mitigation pathway has India warming through mid-century because of committed warming. A "target" showing Kolkata 1.2 °C *cooler* than today is a mitigation aspiration drawn on a physical-temperature axis, and it reads as a forecast.

**Fixed:** the negative pathway is **deleted**. The remaining two are cited to Dhara et al. 2025, *PLOS Climate* (post-AR6 India update): **SSP2-4.5 +1.25 °C** (2041–2060) and **SSP5-8.5 +4.1 °C**.

### Also corrected: the score weights

The 40/40/20 split had **no published precedent** and implied greening and cooling were each exactly twice as important as cost. Equal weighting is the most commonly applied approach in composite-indicator practice (OECD/JRC 2008) and the convention in urban-resilience indices. **Now equal thirds**, cited. An arbitrary split dressed as a derived one is worse than a plain one.

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## 5. The validation harness

`scripts/validate-model.mjs` scores model output against cited benchmarks on every run.

| Check | Value | Expected | |
|---|---|---|---|
| Land-cover thermal contrast | 10.35 °C | 8–18 °C | ok |
| Max ward cooling ≤ local park cooling | 4.45 °C | ≤ 5.42 °C | ok |
| Local park cooling | 5.42 °C | 4.83–8.07 °C | ok |
| Vegetated surface below air | 1.61 K | ≤ 4 K | ok |
| Built surface above air | 14.08 K | 5–25 K | ok |
| Facade vs cool-roof cost per m² | 63× | 40–80× | ok |
| Score at zero intervention | 2 | 0–10 | ok |
| Score at full intervention | 66 | 0–100 | ok |

**8 of 8**, from 4 of 10 before the audit.

**One methodological note you should interrogate.** Three checks were demoted from pass/fail to informational. Before doing that I tested whether they were real defects, and all three borrowed **air-temperature** benchmarks for a model computing **surface** temperature. The clearest evidence: WRI India's figures for the *same* street trees are **−0.3 °C air** and **−27.5 °C road surface** — a 90× gap. No published ward-mean *surface* benchmark exists, so max cooling is now bounded by a physically necessary rule instead (a field's mean cannot shift further than its largest local change). "My benchmark was wrong" is what motivated reasoning sounds like, so the reasoning is written into the code for you to check.

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## 5A. Validation against measurement — what it found

This section is new. Everything above it checks the model against *published literature*; this checks it against *satellite observations of our own three wards*.

### The calibration set

| | |
|---|---|
| Source | NASA ECOSTRESS L2T LSTE v002, 70 m land-surface temperature |
| Window | 2024-01-01 → 2026-07-01, both day and night overpasses |
| Attempted | 179 acquisitions |
| Usable | **49 scenes** (31 night), after cloud and coverage screening |
| Excluded | 130, **each recorded with its reason** — no silent exclusions |
| Urban / rural classes | GHS-SMOD R2023A (EU/UN Degree of Urbanisation), not a temperature threshold |
| Meteorology | NASA POWER hourly, matched to each overpass in local solar time |
| Land cover | ESA WorldCover 10 m, measured inside the same masks — not assumed |

Two controls were run rather than assumed. The urban–rural **view-angle** difference correlated with the measured heat-island signal at $r = -0.322$ ($p = 0.024$), meaning about 10 % of the apparent signal was sensor geometry rather than climate; restricting to near-nadir scenes (≤ 0.75°) removes it and retains 36 scenes. The three alternative definitions of "rural" disagree by only 0.08 °C, so that choice is not a material source of uncertainty.

### How accurate the model is

The important question is not only "how close is our model" but "**how close could any model get** with the data available". So alongside the model's own error we computed a **ceiling**: the error of the best possible statistical predictor built from the same weather inputs, scored leave-one-out so it cannot flatter itself.

| | scenes | best achievable | our model | reported |
|---|---|---|---|---|
| Night | 20 | 2.18 °C | 3.13 °C | **± 3.5 °C** |
| Day | 12 | 3.33 °C | 4.61 °C | **± 5.0 °C** |

Read the "best achievable" column first. **At night the model has real headroom** — a better model could reach ~2.2 °C, so continued work is worthwhile. **By day, no model driven by this weather data can beat ~3.3 °C.** Daytime surface temperature depends on cloud timing, local sunshine and soil moisture that a 50 km reanalysis grid cell simply cannot see. That is a property of the input data, not a defect we can engineer away, and the only real fix is higher-resolution meteorology.

This is why the tool now labels the daytime view **indicative only** and reserves quantitative language for night. Presenting a ± 5 °C daytime figure as decision-grade would be the actual inaccuracy.

### What those figures do NOT cover — spatial skill

Everything in the table above is a **ward-mean** error: how far the model's average temperature sits from ECOSTRESS's average over the same 1400 m box. It says nothing about whether the pattern *inside* the ward is right — and the pattern is what a reader looking at a map actually uses. A model that got the ward average exactly right while placing every hot block in the wrong street would score identically in that table.

So we measured the other thing, at ECOSTRESS's native 70 m, with the ward mean removed from both sides so this is pattern only. 81 ward-scenes across 32 near-nadir scenes and three wards:

| predictor | overall r | day | night |
|---|---|---|---|
| **Our model** | **0.113** | 0.175 | 0.074 |
| Built fraction alone | 0.037 | -0.012 | 0.067 |
| **Vegetation alone** | **0.229** | 0.297 | 0.186 |

**A plain vegetation map predicts within-ward heat about twice as well as our full physics model.** The two null predictors are the reason that is legible at all: $r = 0.11$ on its own reads as "some skill", and only the comparison shows it is worse than a single one of the layers the model is built from. Publishing the correlation without the nulls would have been technically true and materially misleading.

Decomposing the field into its three spatially-varying terms says why:

| term | correlation with measured heat | spatial variation it contributes |
|---|---|---|
| Solar / albedo | 0.040 | 0.02 °C |
| **Built fraction** | **0.037** | **1.46 °C** |
| Vegetation | 0.229 | 0.53 °C |

The built-fraction term contributes most of the picture and carries no measurable skill; the vegetation term carries the skill and is outweighed roughly three to one. So the map's block-by-block detail is largely a rendering of building-footprint density, and footprint density does not predict where ECOSTRESS sees heat at 70 m. The likely reason is physical rather than a coding error: in a dense ward nearly every cell is built, and daytime surface temperature is set by roof material and shading, which a footprint polygon does not carry.

**This is the second independent line of evidence pointing at the model's structure rather than its constants.** The fit above ran all three free parameters to their bounds; this shows the term that dominates the spatial field has no skill. Both point the same way, and together they say that tuning constants is not where the remaining error is.

**What this does not affect.** Ward-level accuracy is measured separately and is unchanged, as are the DC-URS resilience scores — the score reads ward-level indicators and never the per-cell field. The finding bears on one claim only: that the hot and cool blocks *within* a ward mean something.

**What we did about it.** The tool now states it. The colour legend carries the measured figure, the honesty banner says "block detail illustrative" at every screen width, and both readout tooltips carry the full sentence including this vegetation comparison. The figures are held as data in `src/scripts/climate-engine/accuracy.ts` and asserted, so the wording cannot go on claiming the pattern is unvalidated after that stops being true.

Method and its limits, for the record: near-nadir scenes only; cells dropped where ECOSTRESS is cloudy, low-quality or water; the model evaluated at equilibrium with no lateral diffusion, so real advection between adjacent cells counts against it; a ward is about 21 × 21 cells at 70 m, so a single scene's correlation is noisy and the aggregate is the figure. Reproduce with `python3 scripts/measure-spatial-accuracy.py`; full per-scene output in `data/calibration/spatial-accuracy.json`.

### The fit failed, and that is the honest result

We then attempted to fit the model's three remaining free constants — anthropogenic heat, the radiative-to-convective coupling ratio, and the sky-emissivity coefficient — to the measured scenes, with residuals taken on urban *and* rural absolute temperatures rather than on their difference alone. Matching a difference while both sides are wrong is not a calibration.

It did not pass. **All three constants ran to the edge of their physically defensible bounds** and the error stayed at 3.75 °C against a ± 2 °C acceptance bar. Every one moved in the direction that makes the modelled surface warmer, and it was still too cold. Widening the bounds would have produced a number rather than a calibration, so the fitted values were **not adopted** — the model ships with its previous, literature-derived constants, and the fit output is recorded with an explicit "do not ship" flag.

Three candidate explanations were tested and none survived:

| Hypothesis | Result |
| --- | --- |
| Missing heat-storage / thermal inertia | Error moved only 3.75 → 3.41 °C. Insufficient. |
| Evapotranspiration limited by soil moisture rather than air dryness | Removed the bias but *worsened* the scatter; the parameter ran to its bound, so it merely reduced evaporation overall. |
| Vegetation vigour (NDVI) explaining the daytime signal | Ruled out earlier — rural vegetation index exceeds urban in every scene regardless of the sign of the measured heat island. |

What the evidence does point to is that **the evapotranspiration coefficient is too strong** — the one set in §4.2 from local park-cooling measurements. Regional satellite means disagree with local park measurements, which are genuinely different quantities (a park interior versus a mask-wide average). That tension is unresolved and is listed in §6 rather than split down the middle.

### Why the acceptance bar itself was wrong

The ± 2 °C bar was written before anyone knew what was achievable. The ceiling analysis shows it was never reachable for daytime and sits at the edge of possible for night. A target that no model can meet is not a quality standard; the per-phase figures above replace it.

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## 6. What I still don't trust

- **The model is now validated against measurement, and it did not pass cleanly.** This replaces the previous version's statement that nothing had been checked against our own wards. It has been (§5A), and the result is ± 3.5 °C at night and ± 5.0 °C by day — usable for screening and ranking interventions, not for anything requiring a defensible absolute temperature. The daytime figure cannot be improved without better weather data.
- **The map's within-ward pattern is not validated, and is weaker than a vegetation map.** Measured against ECOSTRESS at 70 m (§5A), our field scores r = 0.11 where vegetation cover alone scores 0.23. The built-fraction term contributes 1.46 °C of spatial variation against vegetation's 0.53 °C while correlating at 0.037, so the block-level detail is mostly footprint density rendered as heat. It is disclosed in the product. Ward-level figures and the DC-URS scores are unaffected. Fixing it is a change to the model's structure, not its constants — which is the same conclusion the failed fit reached independently.

- **The evapotranspiration coefficient is contested by our own measurements.** §4.2 derives L = 0.43 from local park-cooling studies and it satisfies those constraints. The satellite fit says it is too strong. These measure different things — a park interior versus a ward-wide average — so this is a real conflict between two valid datasets, not an error to be averaged away. It needs a deliberate decision.
- **The greening benchmark is transplanted.** Berlin's 0.30–0.60 band applies to *individual development sites in a temperate European city*. We apply it to *whole wards in tropical Kolkata*. Convenient, not validated.
- **Efficiency is still tool-relative.** No published cost-per-degree-cooled benchmark exists, so a ward can only score well relative to what our own model produces. Now labelled as such rather than presented as grounded.
- **No Indian wetland cost figure exists.** Government schemes fund *management*, not per-hectare creation. Also worth knowing: East Kolkata Wetlands sit under the 2006 Conservation Act, which prohibits conversion — so a "create wetland" lever may be modelling a legally impossible action. The lever is currently retired from the interface.
- **Several citations were verified from abstracts only.** Re-verification before any client deliverable has not happened.
- **A licensing question for you:** our building footprints are ODbL, which carries share-alike obligations for derived *databases*. Rendered maps are almost certainly fine; the derived data files we ship may not be. Worth a proper answer before anything paid.

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## 7. Open decision

Whether the Green Score should retain the **cooling term**, now that we know how uncertain it is.

The previous version of this document posed this question while the accuracy was unknown. It is now measured: ± 3.5 °C at night, ± 5.0 °C by day. That changes the question from "*can we trust it?*" to "*is this precision good enough for what we sell?*"

A score built only on the greening ratio and cost would be narrower but fully defensible today — both rest on published standards and cited Indian unit costs. Including modelled cooling makes the score more useful and less certain, and it is currently two of the three components (cooling directly, plus efficiency, which is cooling divided by cost).

Two things worth weighing. In its favour: the score uses cooling as a **relative** quantity — ranking interventions against each other under identical forcing — and much of the absolute error is common to both sides of that comparison, so the ranking is more reliable than the ± figure suggests. Against it: we cannot currently prove that, and a client is entitled to ask.

A middle option exists that was not available before: retain the cooling term for the **night** phase, where the model is calibrated, and treat the daytime contribution as indicative — matching what the interface already tells users.

This is a judgement about what the firm is willing to stand behind rather than a technical question, so it sits with the partners rather than with engineering.

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## 8. Sources

Verification status is marked per item: **[V]** full text or primary document consulted · **[A]** abstract or catalogue record only, not yet fully verified.

### Scoring standards and index methodology

| Source | Used for | Link |
|---|---|---|
| Berlin **Biotope Area Factor** (Senate Dept. for Urban Development) **[V]** | Greening weights; 0.30–0.60 target band | berlin.de/sen/uvk/en/nature-and-green/landscape-planning/baf-biotope-area-factor |
| **Seattle Green Factor** (SDCI) **[V]** | Cross-check on surface weights | seattle.gov/sdci/codes/codes-we-enforce-(a-z)/seattle-green-factor |
| Singapore **Green Plot Ratio** (URA LUSH) **[A]** | Third comparator on the weighted-area form | ura.gov.sg/Corporate/Guidelines/Development-Control/Non-Residential/EI/Greenery |
| OECD/JRC (2008), *Handbook on Constructing Composite Indicators* **[V]** | Equal weighting; sensitivity-analysis requirement | doi:10.1787/9789264043466-en |

### Physics, cooling evidence, and the surface-vs-air distinction

| Source | Used for | Link |
|---|---|---|
| Voogt & Oke (2003), *Thermal remote sensing of urban climates*, **RSE** 86(3):370–384 **[V]** | **§4.1** — daytime *surface* UHI 10–15 °C vs canopy *air* UHI 2–5 °C | doi:10.1016/S0034-4257(03)00079-8 |
| Mitra et al. (2022), **Frontiers in Environmental Science** — tropical megacities incl. Kolkata **[V]** | **§4.2** — park cool-island 4.83–8.07 °C; TVoE 0.77 ha; reach 420 m | doi:10.3389/fenvs.2022.1073914 |
| Gunawardena & Steemers (2023), *Neighbourhood-scale vertical greening*, **Buildings & Cities** 4(1) **[V]** | **§4.3** — heat-island intensity 1.86→1.81 K (~3%); energy 2.1–5.2% | doi:10.5334/bc.282 |
| **LBNL Heat Island Group** — cool roof materials **[V]** | Albedo 0.15 dark / 0.60 aged-cool | heatisland.lbl.gov/coolscience/cool-roofs |
| **WRI India**, *Urban Trees' Cooling Potential* **[V]** | Canopy dose-response; the −0.3 °C air vs −27.5 °C surface contrast in §5 | wri.org/insights/urban-trees-cooling-potential |

## Climate projections

| Source | Used for | Link |
| --- | --- | --- |
| Dhara, Deshpande, Roxy, Dalpadado & Shrestha (2025), **PLOS Climate** 4(11):e0000724 **[V]** | **§4.5** — SSP2-4.5 +1.25 °C (2041–60); SSP5-8.5 +4.1 °C | doi:10.1371/journal.pclm.0000724 |
| Ministry of Earth Sciences (2020), *Assessment of Climate Change over the Indian Region* **[A]** | Cross-check: RCP4.5 +2.41 ± 0.40 °C by 2066–95 | link.springer.com/book/10.1007/978-981-15-4327-2 |

## Urban heat island benchmarks (used to sanity-check, not to calibrate)

| Source | Kolkata / India value | Link |
| --- | --- | --- |
| Nayak, Vinod & Prasad (2023), **Applied Sciences** 13(24):13323 **[A]** | Kolkata night SUHII 0.85 °C annual | doi:10.3390/app132413323 |
| Jain (2023), **Frontiers in Sustainable Cities** 5:1084573 **[A]** | Kolkata night 1.3–1.5 °C (DJF) | doi:10.3389/frsc.2023.1084573 |
| Siddiqui et al. (2021), **Sustainable Cities and Society** 75:103374 **[A]** | Indian metros night 1.34–2.07 °C | doi:10.1016/j.scs.2021.103374 |
| Peng et al. (2012), **Environmental Science & Technology** 46(2):696–703 **[A]** | Global 1.5 °C day / 1.1 °C night; equal-area rural reference method | doi:10.1021/es2030438 |
| Chakraborty & Lee (2019), **Int. J. Applied Earth Obs. & Geoinformation** 74:269–280 **[A]** | Global 0.85 day / 0.55 night; water-masking rationale | doi:10.1016/j.jag.2018.09.015 |
| Shastri et al. (2017), **Scientific Reports** 7:40178 **[V]** | Indian daytime *cool* island in pre-monsoon — why Kolkata needs explaining | doi:10.1038/srep40178 |
| Kumar et al. (2017), **Scientific Reports** 7:14054 **[A]** | 60% of 89 Indian urban areas show a daytime cool island | doi:10.1038/s41598-017-14213-2 |

## Cost sources (all Indian)

| Source | Used for | Link |
| --- | --- | --- |
| **Telangana Cool Roof Policy 2023–2028 [V]** | ₹150/m² cool roof; indoor −2.1 to −4.3 °C | telangana.gov.in (policy PDF) |
| **NRDC**, *Keeping It Cool: How India Can Protect Its People with Cool Roofs* **[V]** | Ahmedabad lime-wash rate; surface up to −30 °C | nrdc.org/resources/keeping-it-cool-how-india-can-protect-its-people-cool-roofs |
| **Gujarat AMRUT 2.0** urban gardens (2026) **[A]** | ₹1.5 crore/ha park capex, from two named projects | Press release; see `heat-map-intervention-model.md` §9 |
| **PreventionWeb / NRDC** — Ahmedabad Heat Action Plan **[A]** | >1,100 deaths avoided/yr; −27% mortality on hottest days | preventionweb.net |

### Input data and licences

| Dataset | Licence | Link |
| --- | --- | --- |
| Microsoft ML Building Footprints | **ODbL** — share-alike, see §6 | github.com/microsoft/GlobalMLBuildingFootprints |
| Google Open Buildings 2.5D | CC BY 4.0 | sites.research.google/open-buildings |
| OpenStreetMap road centrelines | ODbL | openstreetmap.org/copyright |
| Norwegian Meteorological Institute (live air temp) | Free, keyless | api.met.no |
| NASA **ECOSTRESS** L2T LSTE v002 | US public domain | doi:10.5067/ECOSTRESS/ECO_L2T_LSTE.002 |
| **ESA WorldCover** 10 m v200 | CC BY 4.0 | doi:10.5281/zenodo.7254221 |
| **JRC GHS-SMOD** R2023A (Degree of Urbanisation) | CC BY 4.0 | doi:10.2905/A0DF7A6F-49DE-46EA-9BDE-563437A6E2BA |
| **NASA POWER** hourly meteorology (MERRA-2) | US public domain, no key | power.larc.nasa.gov |
| Open-Meteo historical archive | **Non-commercial — rejected** | open-meteo.com/en/license |

**Attribution strings required by licence** are recorded in `docs/heat-map-intervention-model.md` and rendered on the tool itself.

### Methods used in the validation (§5A)

| Source | Used for | Link |
| --- | --- | --- |
| Brutsaert (1975), *Water Resources Research* 11(5):742–744 **[V]** | Clear-sky emissivity, replacing two hard-coded sky temperatures | doi:10.1029/WR011i005p00742 |
| Peng et al. (2012), *Environ. Sci. Technol.* 46(2) **[A]** | Equal-area rural reference method for the heat-island difference | doi:10.1021/es2030438 |
| Voogt & Oke (2003), *Remote Sensing of Environment* 86(3) **[V]** | Surface vs air heat island are different quantities; view-angle sensitivity of thermal remote sensing | doi:10.1016/S0034-4257(03)00079-8 |
| Spencer (1971), *Search* 2(5):172 **[A]** | Solar declination series, for per-scene sun angle | — |
| LP DAAC, *ECOSTRESS L2T LSTE v002 known issues* **[V]** | Quality-flag caveat: two documented QC bits are unset in v002 | lpdaac.usgs.gov |

**[V]** verified against full text · **[A]** verified from abstract or documentation only — see §6.

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Full formulas and every constant: `docs/heat-map-intervention-model.md` Validation harness: `scripts/validate-model.mjs`

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*This document leads with limitations rather than capabilities by design. Anything that reads as understated should be queried — better to over-disclose internally than to have a municipal client discover it first.*